

ECE 313: Problem Set 5: Problems and Solutions

Due: Sunday, March 9 at 11:59 p.m.

Reading: *ECE 313 Course Notes*, Sections 2.11 - 2.12, 3.1 - 3.2

Note on reading: For most sections of the course notes there are short answer questions at the end of the chapter. We recommend that after reading each section you try answering the short answer questions. Do not hand these in; answers to the short answer questions are provided in the appendix of the notes.

Note on turning in homework: Homework is assigned on a weekly basis on Fridays, and is due by 7 p.m. on the following Friday. **This homework is instead due on Sunday, March 9, at 11:59 p.m.** You must upload handwritten homework to Gradescope. Alternatively, you can typeset the homework in LaTeX. However, no additional credit will be awarded to typeset submissions. No late homework will be accepted.

Please write on the top right corner of the first page:

NAME

NETID

SECTION

PROBLEM SET #

Page numbers are encouraged but not required. Five points will be deducted for improper headings. Please assign your uploaded pages to their respective question numbers while submitting your homework on Gradescope. **5 points will be deducted for incorrectly assigned page numbers.**

1. [Geometric Detection]

A server chooses one of two routes, Route 0 or Route 1, and repeatedly transmits a packet over the chosen route until the packet is received without error at the client. X denotes the number of times the packet is transmitted over the chosen route, including the final error-free transmission.

Assuming that the successive transmissions are independent trials of an experiment, the two hypotheses are

- H_0 : Route 0 is used for packet transmission: $X \sim \text{Geo}(p_0)$
- H_1 : Route 1 is used for packet transmission: $X \sim \text{Geo}(p_1)$

where p_0 and p_1 are the respective probabilities of error-free transmission, with $0 < p_0 < p_1 < 1$.

- (a) State the maximum likelihood decision rule.

Solution:

$$\Lambda(k) = \frac{(1 - p_1)^{k-1} p_1}{(1 - p_0)^{k-1} p_0}$$

When $\Lambda(k) \geq 1$, we declare H_1 ; otherwise, we declare H_0 .

- (b) Suppose the transmitter chooses Route 0 and Route 1 with probabilities π_0 and $\pi_1 = 1 - \pi_0$, respectively. For what values of π_0 does the MAP decision rule always choose H_0 ? For what values of π_1 does the MAP decision rule always choose H_1 ?

Solution: For the $\Lambda(k)$ we found earlier, we know that $1 > p_1 > p_0 > 0$ and therefore $0 < 1 - p_1 < 1 - p_0 < 1$. As a result, $(\frac{1-p_1}{1-p_0})^{k-1}$ monotonically decreases on k , and therefore $\Lambda(1)$ is the largest possible value of $\Lambda(k)$ and $\lim_{k \rightarrow \infty} \Lambda(k)$ goes to 0.

H_0 : If we can show that $\Lambda(1) < \frac{\pi_0}{1-\pi_0}$, then we also know that $\Lambda(k) \leq \Lambda(1) < \frac{\pi_0}{1-\pi_0}$ for all k .

$$\begin{aligned}\frac{\pi_0}{1-\pi_0} &> \Lambda(1) = \frac{p_1}{p_0} \\ \implies \pi_0 \cdot p_0 &> p_1 \cdot (1-\pi_0) \\ \implies \pi_0 &> \frac{p_1}{p_0+p_1}\end{aligned}$$

H_1 : Only when $\pi_0 = 0$ is $\Lambda(k)$ guaranteed to be larger than $\frac{\pi_0}{\pi_1}$ for all k . So when $\pi_0 = 0$, the MAP decision rule chooses H_1 .

- (c) Assume $p_0 = 0.5$, $p_1 = 0.7$, $\pi_0 = \frac{1}{3}$, and $\pi_1 = \frac{2}{3}$. Find the error probability for the MAP rule. Break ties in favor of H_1 if necessary.

Solution: Now,

$$\Lambda(k) = \frac{(1-0.7)^{k-1} \cdot 0.7}{(1-0.5)^{k-1} \cdot 0.5} = \left(\frac{3}{5}\right)^{k-1} \cdot \frac{7}{5}$$

We can start by finding the values of k such that we select H_1 .

$$\begin{aligned}\frac{1}{2} &\leq \left(\frac{3}{5}\right)^{k-1} \cdot \frac{7}{5} \\ \implies \ln\left(\frac{5}{14}\right) &\leq (k-1) \ln\left(\frac{3}{5}\right) \\ \implies \frac{\ln\left(\frac{5}{14}\right)}{\ln\left(\frac{3}{5}\right)} &\geq (k-1)\end{aligned}$$

So for $k \in \{1, 2, 3\}$, we select H_1 , and for $k \in \{4, 5, 6, \dots\}$ we select H_0 .

The probability that H_0 is true but we declare H_1 to be true (i.e. $p_{\text{false alarm}}$) is:

$$\begin{aligned}P(\text{declare } H_1 | H_0 \text{ true}) &= P(X \in \{1, 2, 3\} | H_0 \text{ true}) \\ &= P(X = 1 | H_0 \text{ true}) + P(X = 2 | H_0 \text{ true}) + P(X = 3 | H_0 \text{ true}) \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \\ &= \frac{7}{8}\end{aligned}$$

The probability that H_1 is true but we declare H_0 to be true (i.e. p_{miss}) is:

$$\begin{aligned}P(\text{declare } H_0 | H_1 \text{ true}) &= P(X \in \{4, 5, 6, \dots\} | H_1 \text{ true}) \\ &= 1 - P(X \in \{1, 2, 3\} | H_1 \text{ true}) \\ &= 1 - (P(X = 1 | H_1 \text{ true}) + P(X = 2 | H_1 \text{ true}) + P(X = 3 | H_1 \text{ true})) \\ &= 1 - \left(\frac{7}{10} + \frac{7}{10} \cdot \frac{3}{10} + \frac{7}{10} \cdot \left(\frac{3}{10}\right)^2\right) \\ &= \frac{27}{1000}\end{aligned}$$

Finally,

$$p_{\text{error}} = \pi_0 p_{\text{false alarm}} + \pi_1 p_{\text{miss}} = \frac{1}{3} \cdot \frac{7}{8} + \frac{2}{3} \cdot \frac{27}{1000} \approx 0.3097$$

2. [ML and MAP Decision Rules]

Consider a binary hypothesis testing problem, where X has the pmf $p_X(k) = \frac{k}{21}$ for $k = 1, 2, \dots, 6$ under hypothesis H_0 and $X \sim \text{Geo}(\frac{1}{2})$ under hypothesis H_1 .

- (a) State the ML decision rule.

Solution:

$$\Lambda(k) = \frac{1/2^k}{k/21} = \frac{21}{k \cdot 2^k}, \quad k = 1, 2, \dots, 6$$

Therefore, we declare H_1 when $\Lambda(k) \geq 1$, i.e. when $k = 1, 2$, H_0 when $k = 3, 4, 5, 6$. Besides, because the supports are different, we also trivially declare H_1 when $k > 6$.

- (b) Describe the MAP decision rule under the assumption that H_0 is three times as likely as H_1 .

Solution: We are told that $\pi_0 = 3\pi_1$, i.e. $\frac{\pi_0}{\pi_1} = 3$. Thus, we declare H_1 when $\Lambda(k) \geq 3$, i.e. when $k = 1$, H_0 when $k = 2, 3, 4, 5, 6$. Again, we trivially declare H_1 when $k > 6$ for the same reason as in (a).

- (c) Using the prior distribution given in part (b), find the average error probability of the MAP rule.

Solution: We know $\pi_0 + \pi_1 = 1$ and $\pi_0 = 3\pi_1$, so $\pi_0 = \frac{3}{4}$, $\pi_1 = \frac{1}{4}$. Then

$$\begin{aligned} p_e &= \pi_0 p_{\text{false alarm}} + \pi_1 p_{\text{miss}} \\ &= \frac{3}{4} P(X \in \{1, 7, 8, \dots\} | H_0 \text{ true}) + \frac{1}{4} P(X \in \{2, 3, 4, 5, 6\} | H_1 \text{ true}) \\ &= \frac{3}{4} \cdot \left(\frac{k}{21} \Big|_{k=1} + 0 \right) + \frac{1}{4} \cdot \sum_{k=2}^6 \frac{1}{2^k} \\ &= \frac{1}{28} + \frac{1}{16} \cdot \frac{(1 - (1/2)^5)}{1 - 1/2} \\ &= \frac{281}{1792} \approx 0.1568 \end{aligned}$$

3. [CDFs]

For each of the following functions, state whether or not it could be the CDF of some random variable. If so, find the corresponding pmf or pdf. If not, explain why.

- (a)

$$F(c) = \begin{cases} 0 & c \leq 0 \\ 0.5c & 0 < c \leq 1 \\ 0.25 + 0.25c & 1 < c \leq 3 \\ 1 & 3 < c \end{cases}$$

Solution: $F(c)$ is defined on $\mathbb{R}$, being non-decreasing and continuous (and thus right-continuous), and satisfies $\lim_{c \rightarrow -\infty} F(c) = 0$, $\lim_{c \rightarrow \infty} F(c) = 1$. Therefore, it is a valid CDF.

As $F(c)$ is continuous, it corresponds to a pdf $f(x)$, which can be obtained by taking the derivative of the CDF:

$$f(x) = \begin{cases} 0 & x \leq 0 \text{ or } x > 3 \\ 0.5 & 0 < x \leq 1 \\ 0.25 & 1 < x \leq 3 \end{cases}$$

(b)

$$F(c) = \begin{cases} 0 & c \leq 0 \\ 0.5 & 0 < c \leq 1 \\ 0.75 & 1 < c \leq 3 \\ 1 & 3 < c \end{cases}$$

Solution: It is an invalid CDF, since it is not right continuous. For example, $F(1+) = 0.75 \neq 0.5 = F(1)$.

(c)

$$F(c) = \begin{cases} 0.5 & c < 1 \\ 0.75 & 1 \leq c < 3 \\ 1 & 3 \leq c \end{cases}$$

Solution: It is an invalid CDF, since $\lim_{c \rightarrow -\infty} F(c) \neq 0$.

(d)

$$F(c) = \begin{cases} 0 & c < 0 \\ 0.5 & 0 \leq c < 1 \\ 0.75 & 1 \leq c < 3 \\ 1 & 3 \leq c \end{cases}$$

Solution: Similar as in (a), we can verify this is a valid CDF. Since $F(c)$ consists of only horizontal lines, it corresponds to a pmf $p(x)$ which can be obtained by computing ΔF at all jump points:

$$p(x) = \begin{cases} 0.5 & x = 0 \\ 0.25 & x = 1 \\ 0.25 & x = 3 \\ 0 & \text{otherwise} \end{cases}$$

(e)

$$F(c) = \begin{cases} 0 & c < 0 \\ 0.5 & 0 \leq c < 1 \\ 0.75 & 1 \leq c < 3 \\ 1 & 3 < c \end{cases}$$

Solution: It is an invalid CDF, since it is not defined at $c = 3$.

(f)

$$F(c) = \begin{cases} 0.5c & 0 < c \leq 1 \\ 0.25 + 0.25c & 1 < c \leq 3 \\ 1 & 3 < c \end{cases}$$

Solution: It is an invalid CDF, since it is not defined on $c \leq 0$.

4. [Complex CDFs]

Consider the following function:

$$F(u) = \begin{cases} 0 & u < -5 \\ a & u = -5 \\ e^u & -5 < u < -1 \\ b & u = -1 \\ \frac{1}{2} & -1 < u < 1 \\ \frac{1}{8}u + \frac{3}{8} & 1 \leq u < 3 \\ 1 - ce^{-u} & 3 \leq u \end{cases}$$

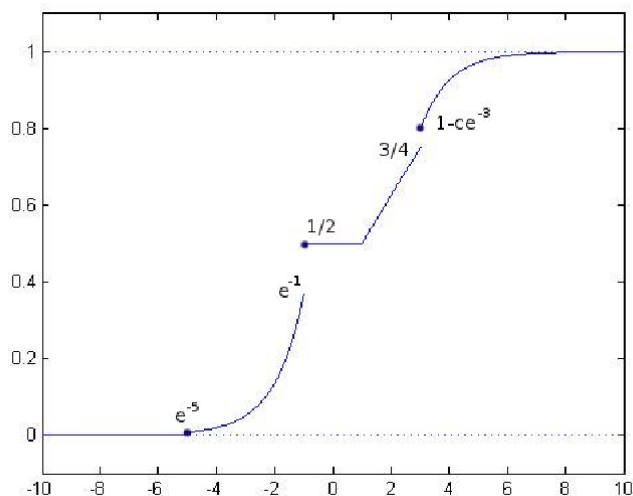
- (a) Obtain the values (or range of values) for a , b , and c that make this a valid CDF.

Solution: To satisfy right continuity, $a = e^u|_{u=-5} = e^{-5}$. To satisfy right continuity, $b = \frac{1}{2}$. c is more complicated.

- c must be nonnegative, because otherwise $1 - ce^{-u} > 1$.
- A CDF must be non-decreasing, so $1 - ce^{-3} \geq \frac{3}{4}$ and therefore $c \leq \frac{e^3}{4}$.

With both of these things in mind, $0 \leq c \leq \frac{e^3}{4}$.

- (b) Sketch $F(u)$ using these values of a , b , and c . If any variable can be in a range of values, pick an arbitrary value.



Solution:

- (c) Now assume that $F(u)$ is the CDF of random variable X . Find the following probabilities:

- $P(X = -5)$
- $P(X = -2)$
- $P(X \leq -5)$
- $P(X \geq -1)$
- $P(X = 0)$
- $P(X = 3)$

Solution:

- $P(X = -5) = F_X(-5^+) - F_X(-5^-) = a - 0 = e^{-5}$

- $P(X = -2) = F_X(-2^+) - F_X(-2^-) = e^{-2} - e^{-2} = 0$
- $P(X \leq -5) = F_X(-5) = a = e^{-5}$
- $P(X \geq -1) = 1 - F_X(-1^-) = 1 - e^{-1}$
- $P(X = 0) = 0$
- $P(X = 3) = 1 - ce^{-3} - \frac{1}{8} \cdot 3 - \frac{3}{8} = \frac{1}{4} - ce^{-3}$